CMOS Medium

Metal oxide Silicon / Semiconductor

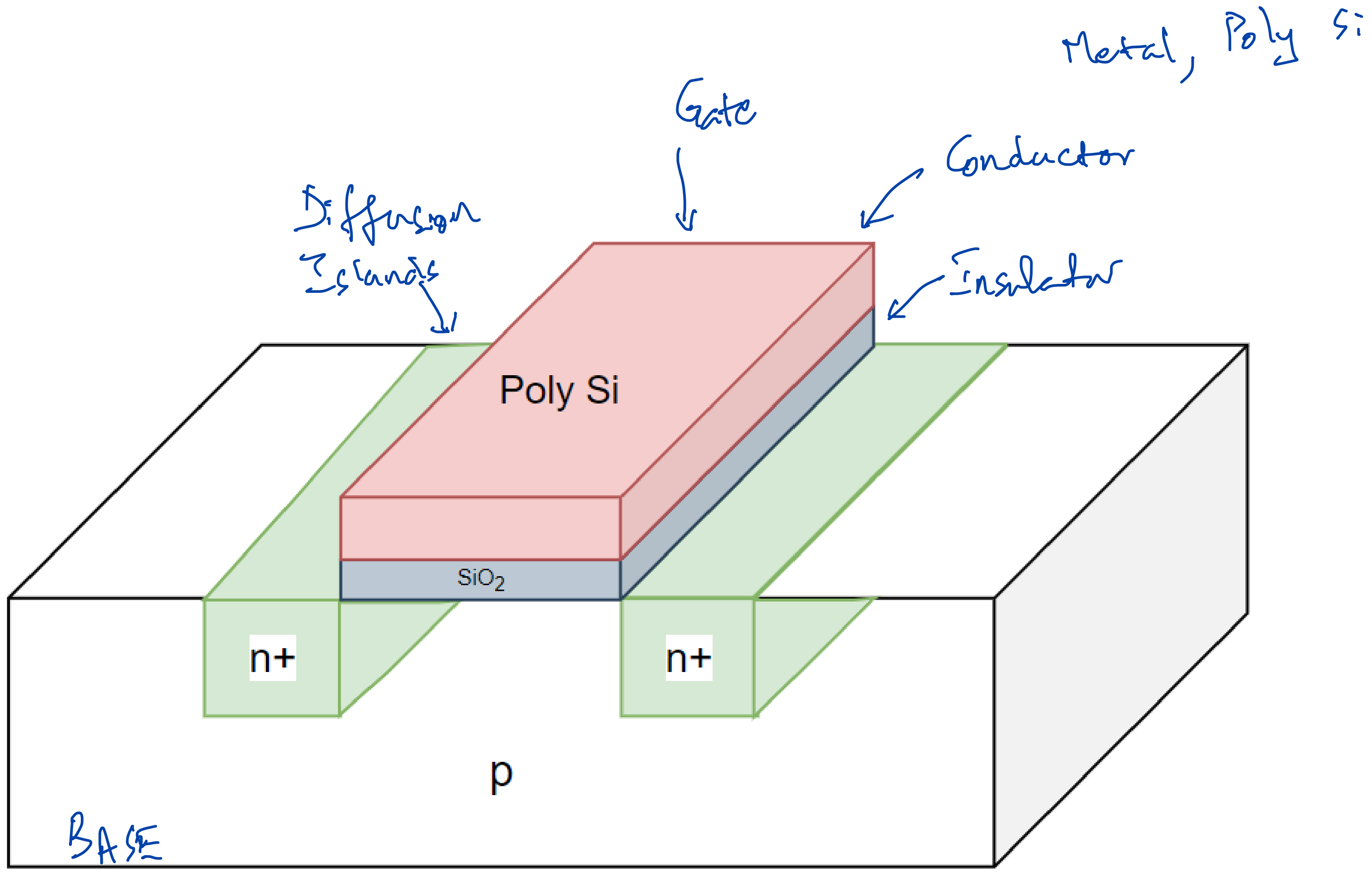
Complementary

Encoding of Logic States

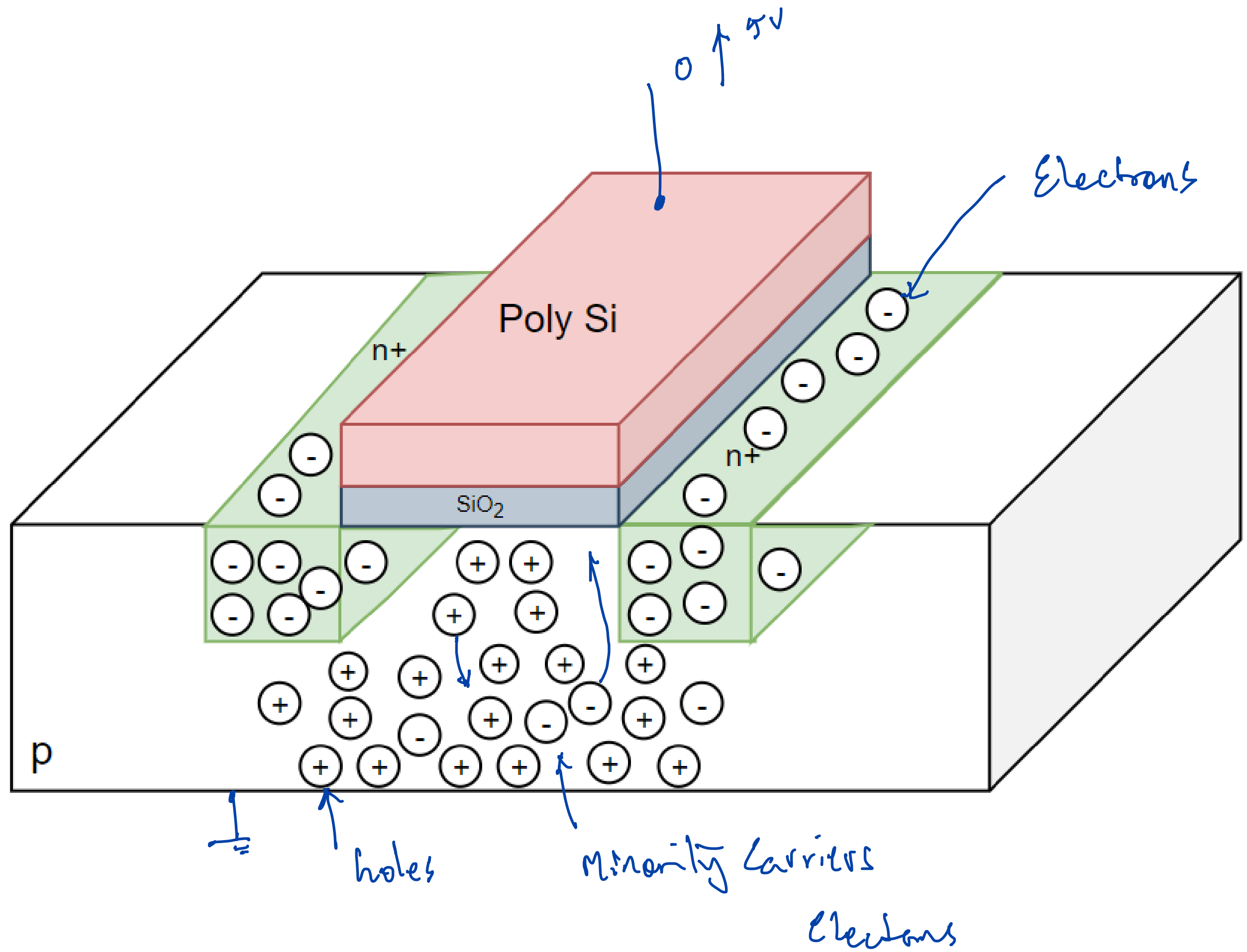
0 | FALSE | LOW — 0V | GND | V_{SS} | V_S

1 | TRUE | HIGH — 5V | V_{DD} | V_D

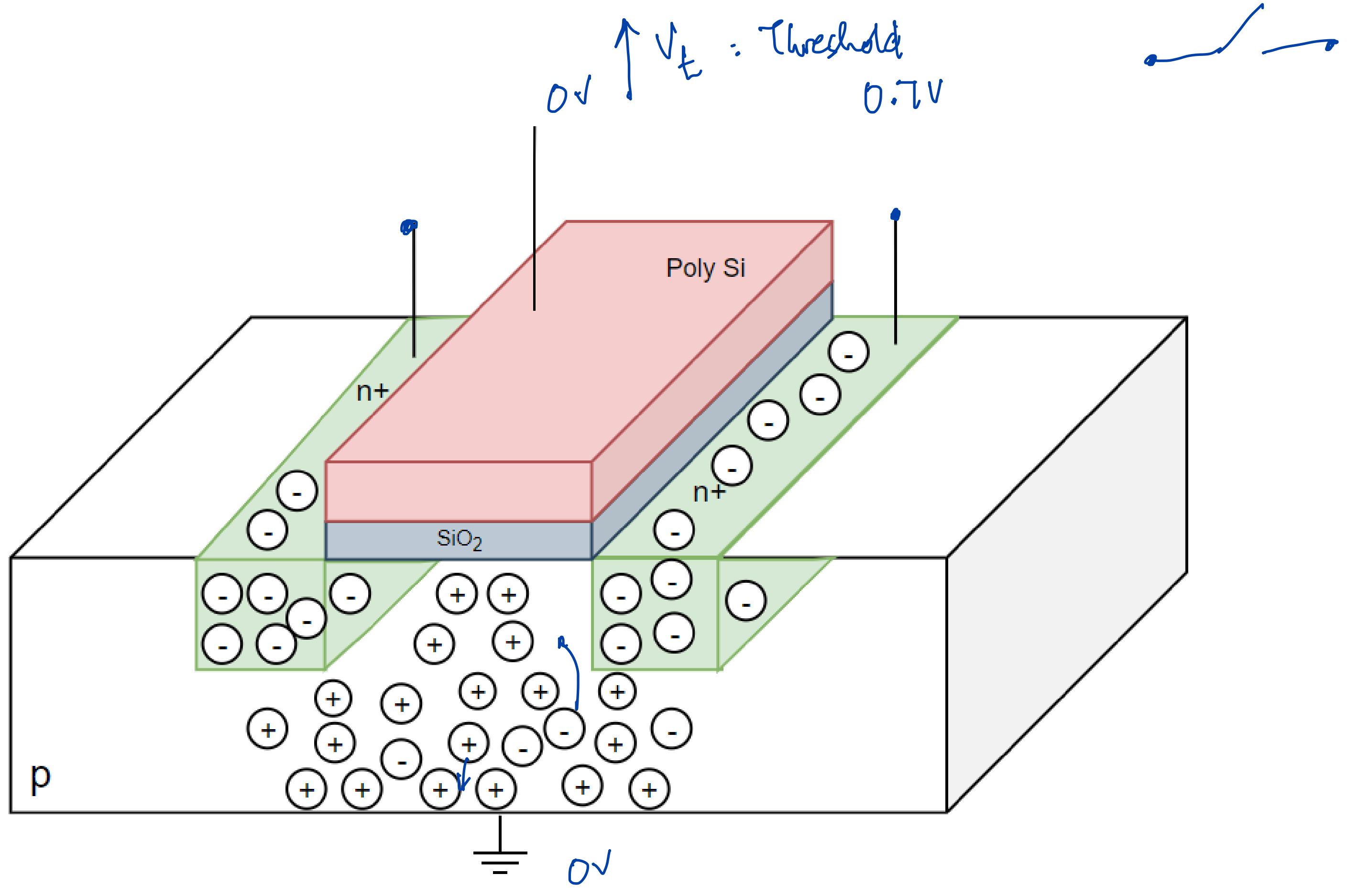
n-transistor



Carriers in n-transistor



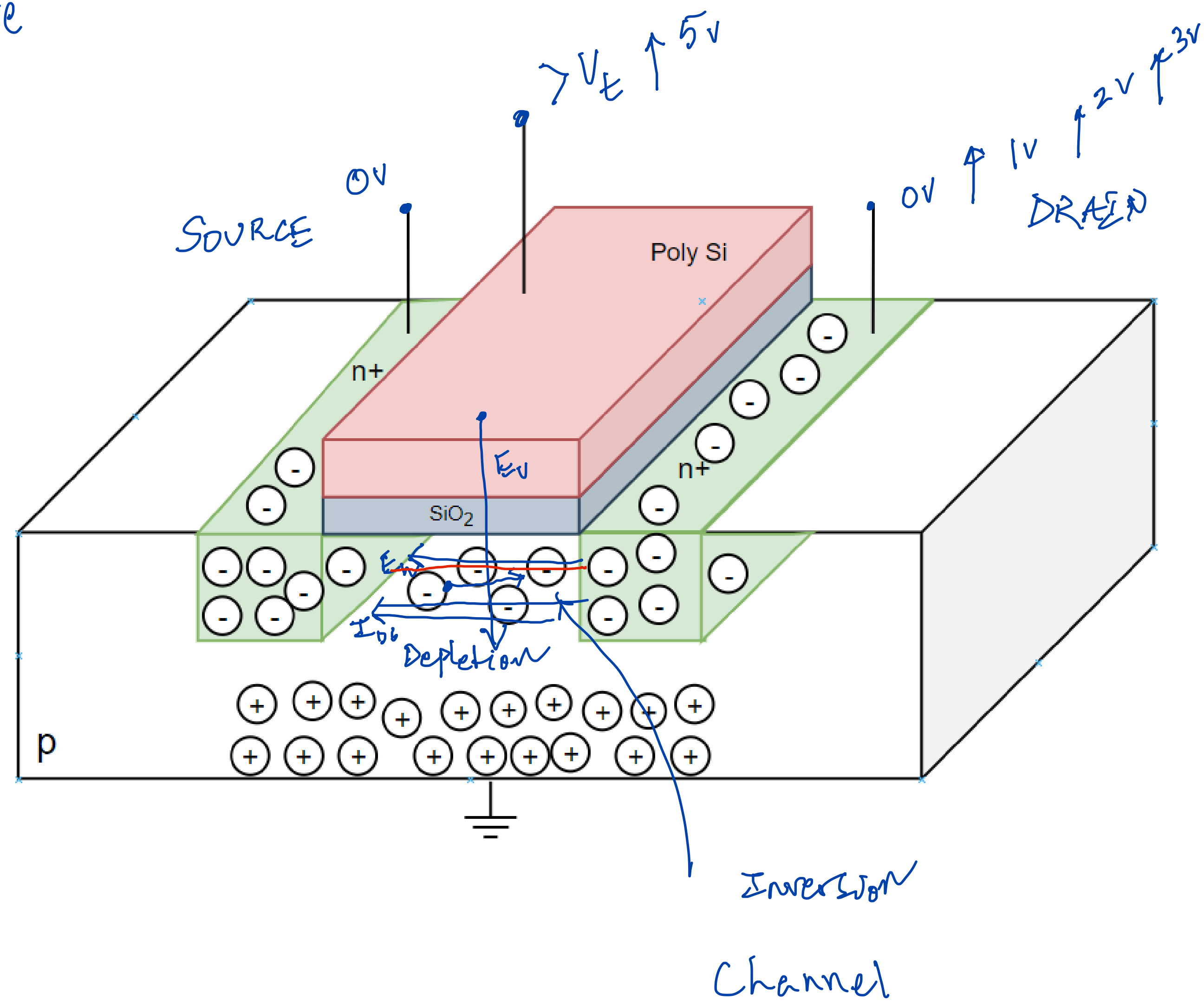
n-transistor as normally-open switch



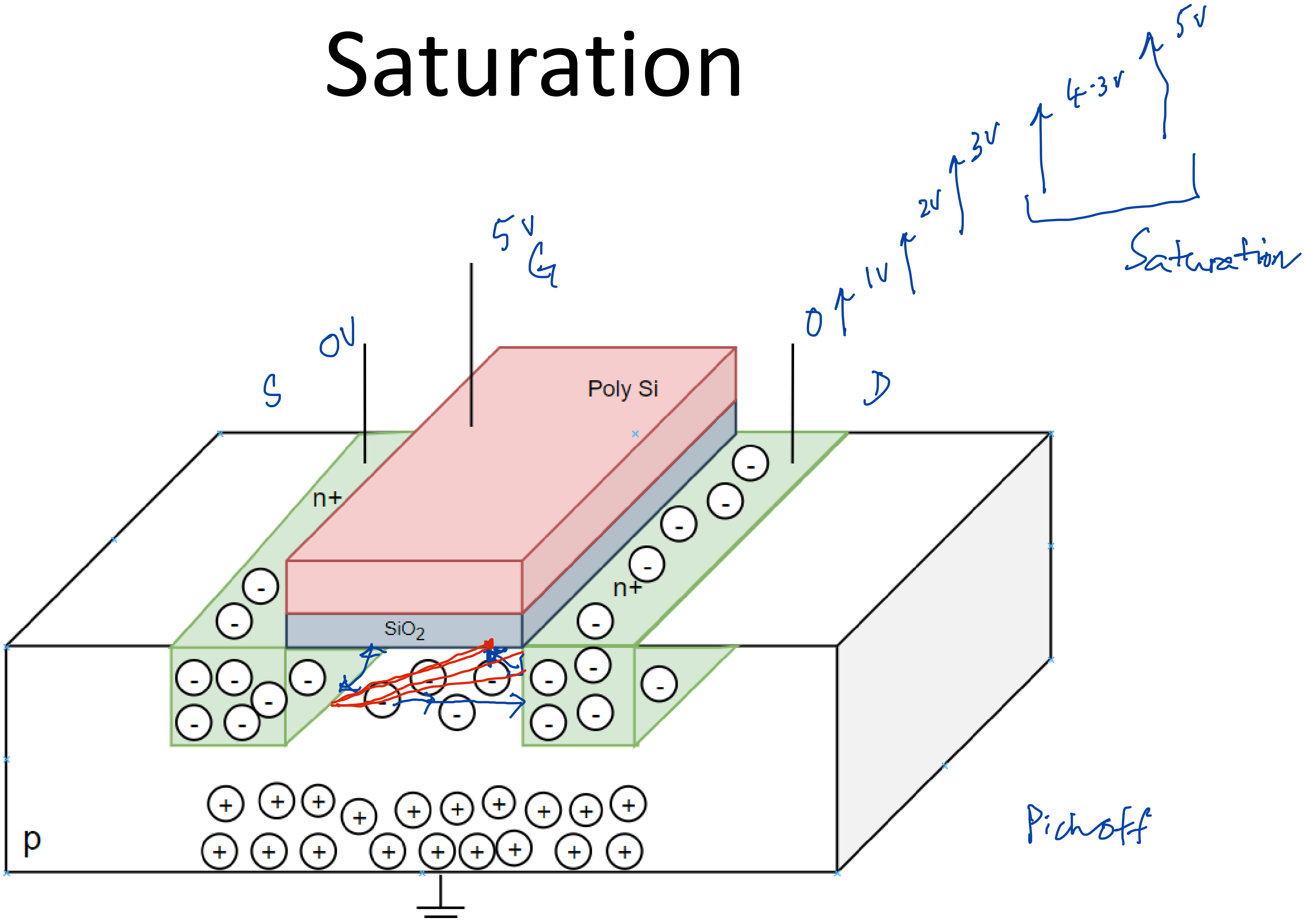
Conduction

Linear
Resistive

Closed Switch

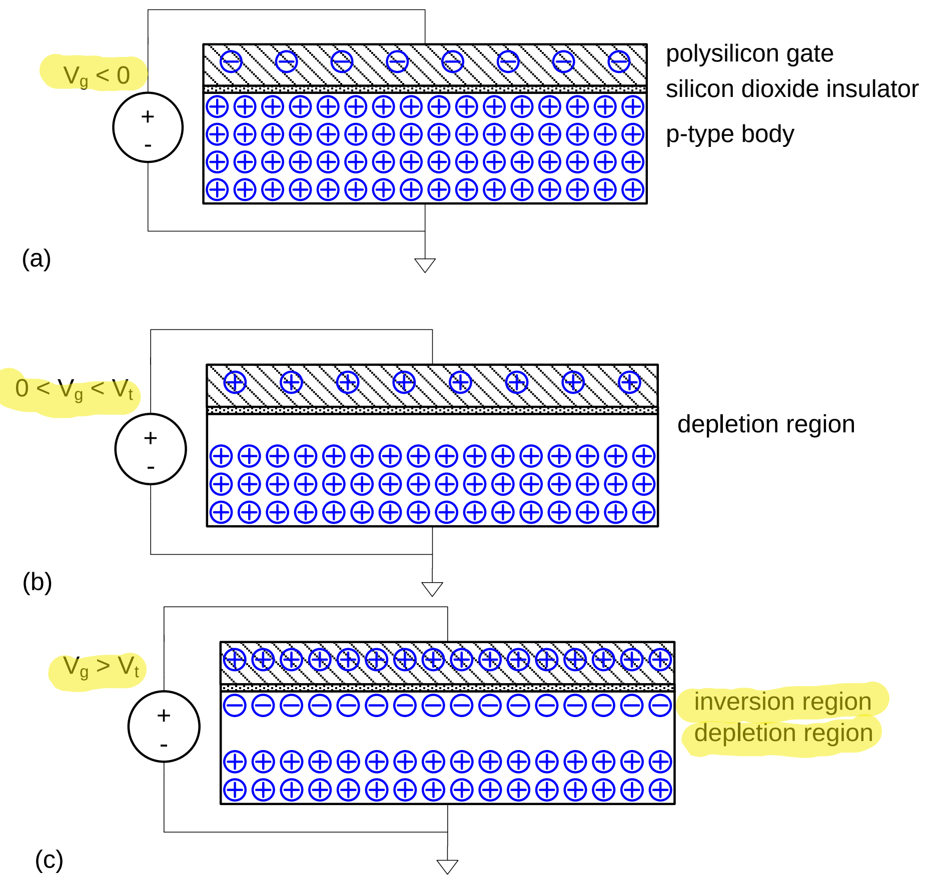


Saturation

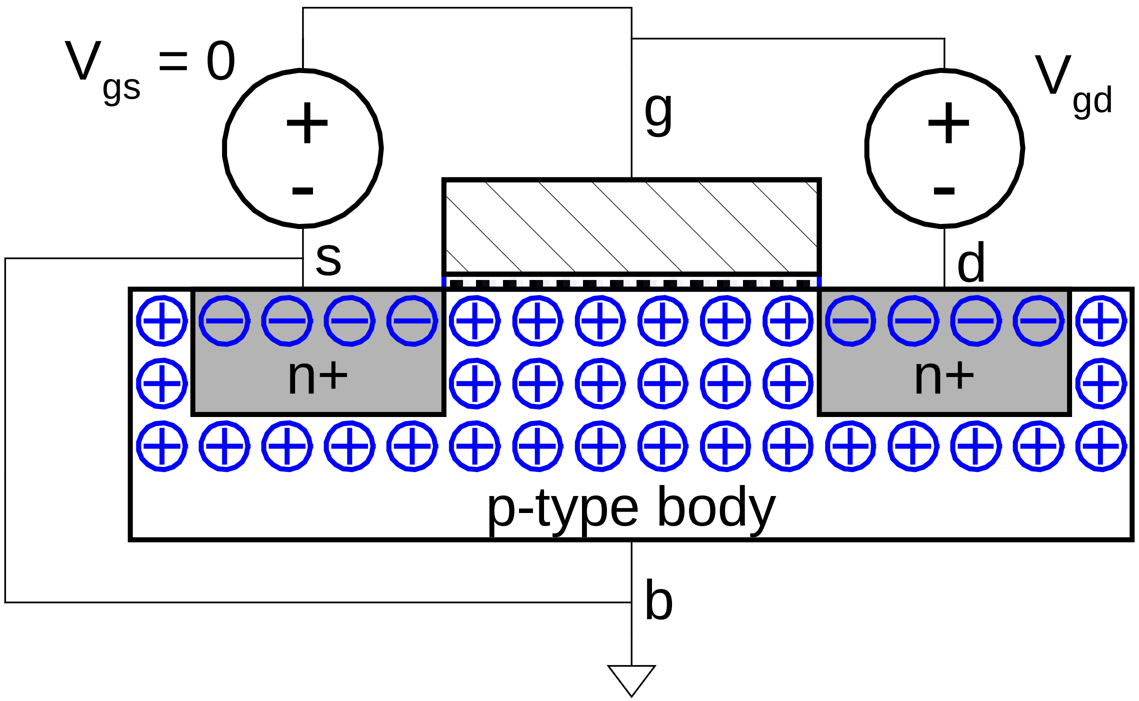


Modes of Operation

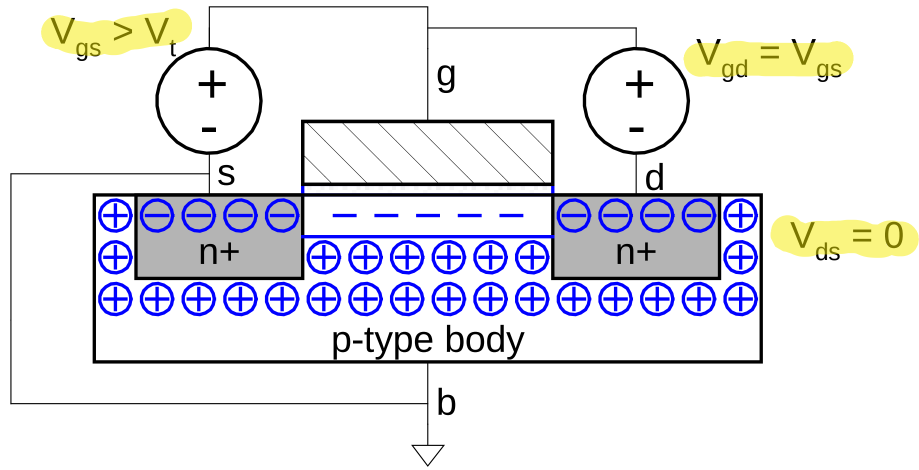
Accumulation
Depletion
Inversion



Cutoff



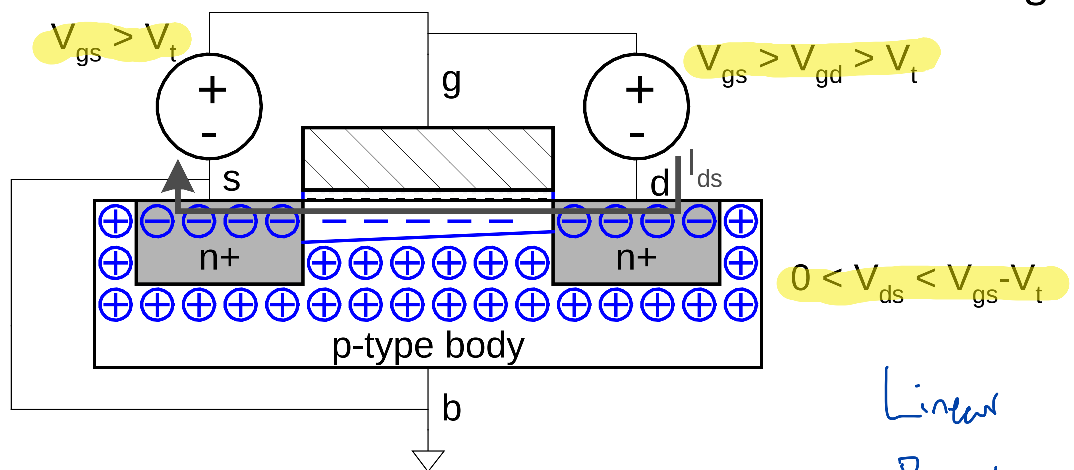
Linear Mode



$V_{ds} = V_{gs} – V_{gd}$

or

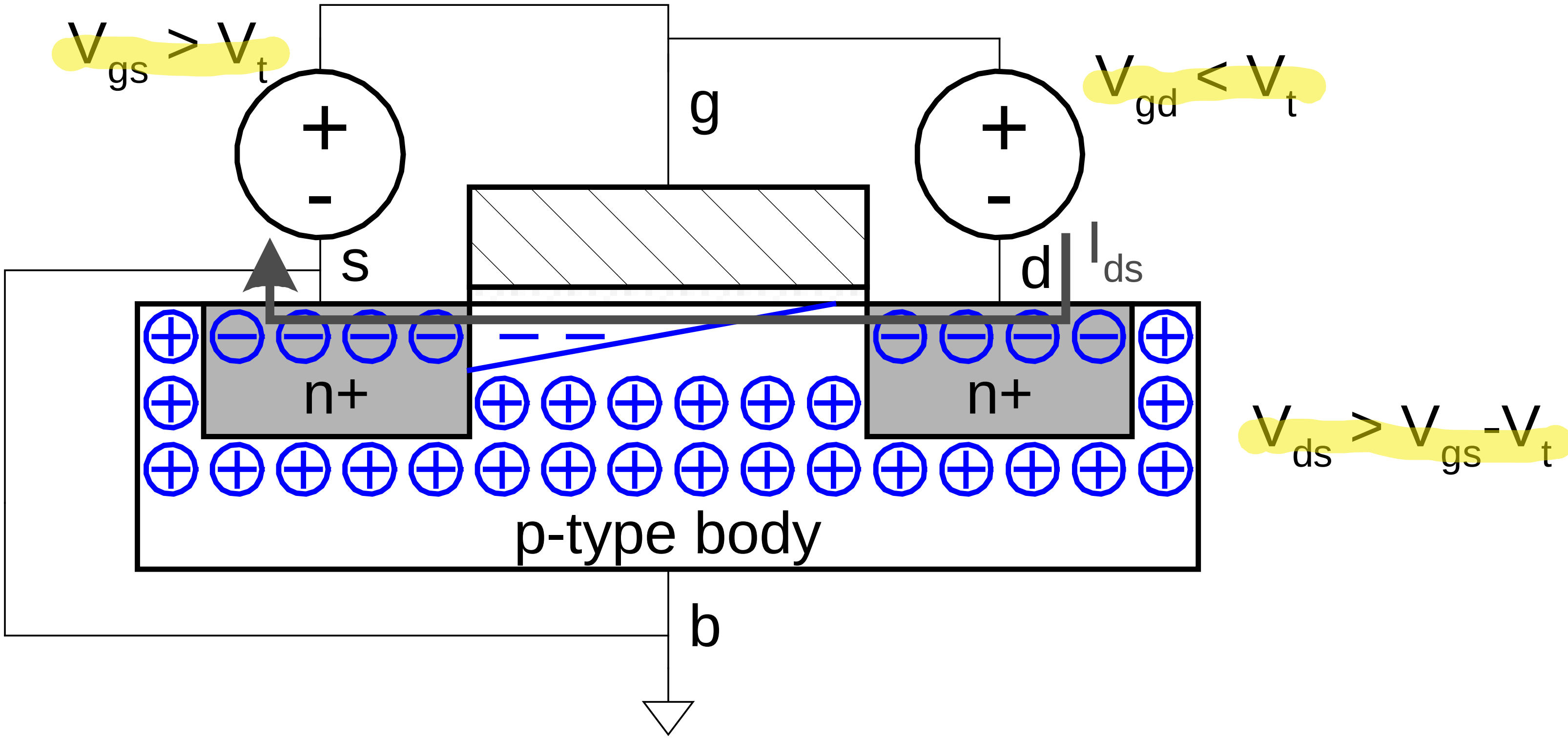
$V_{gd} = V_{gs} - V_{ds}$



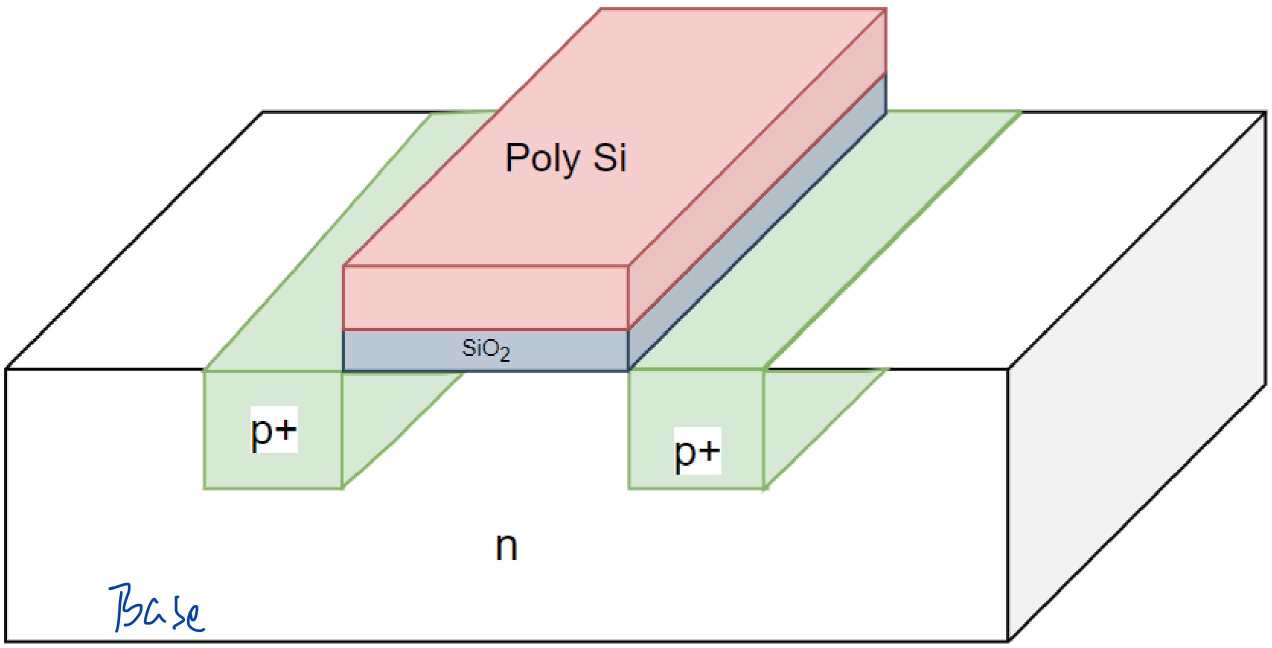
Linear

Resistive

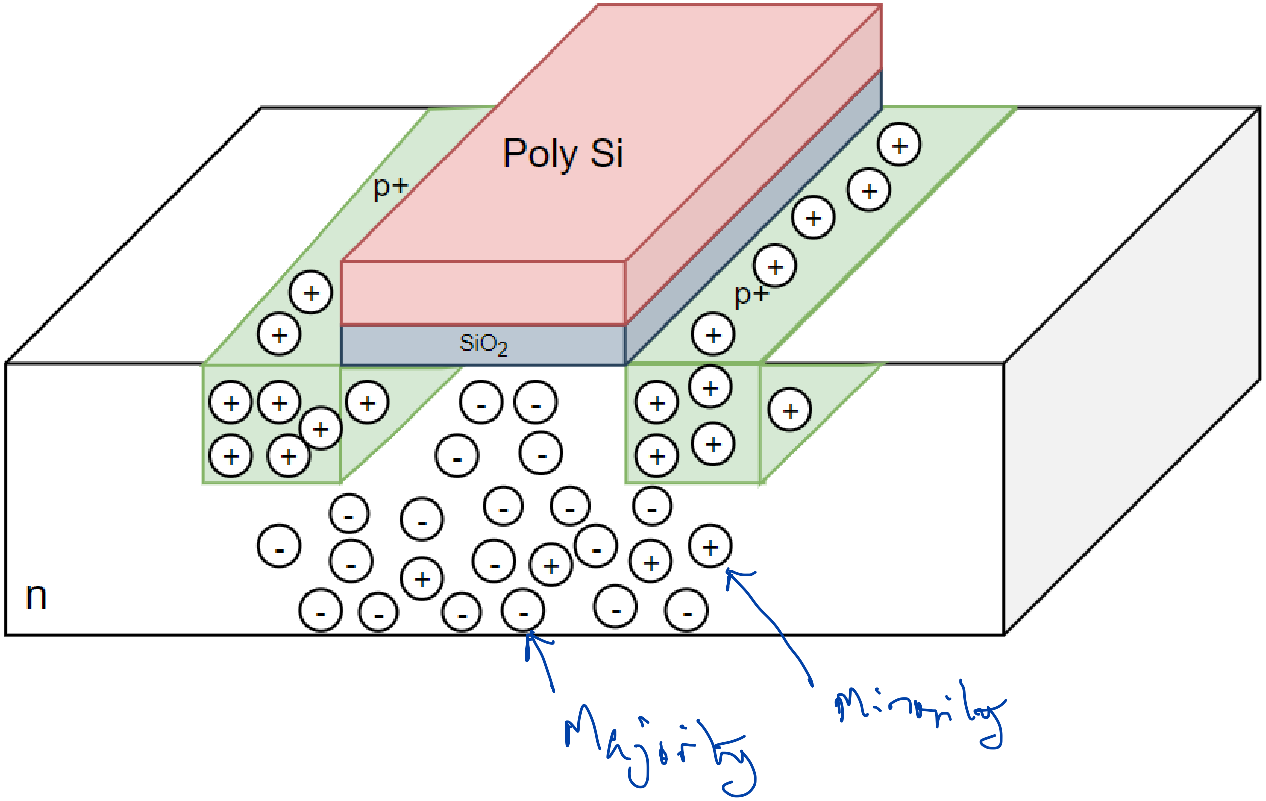
Saturation - Pinch Off



p-transistor

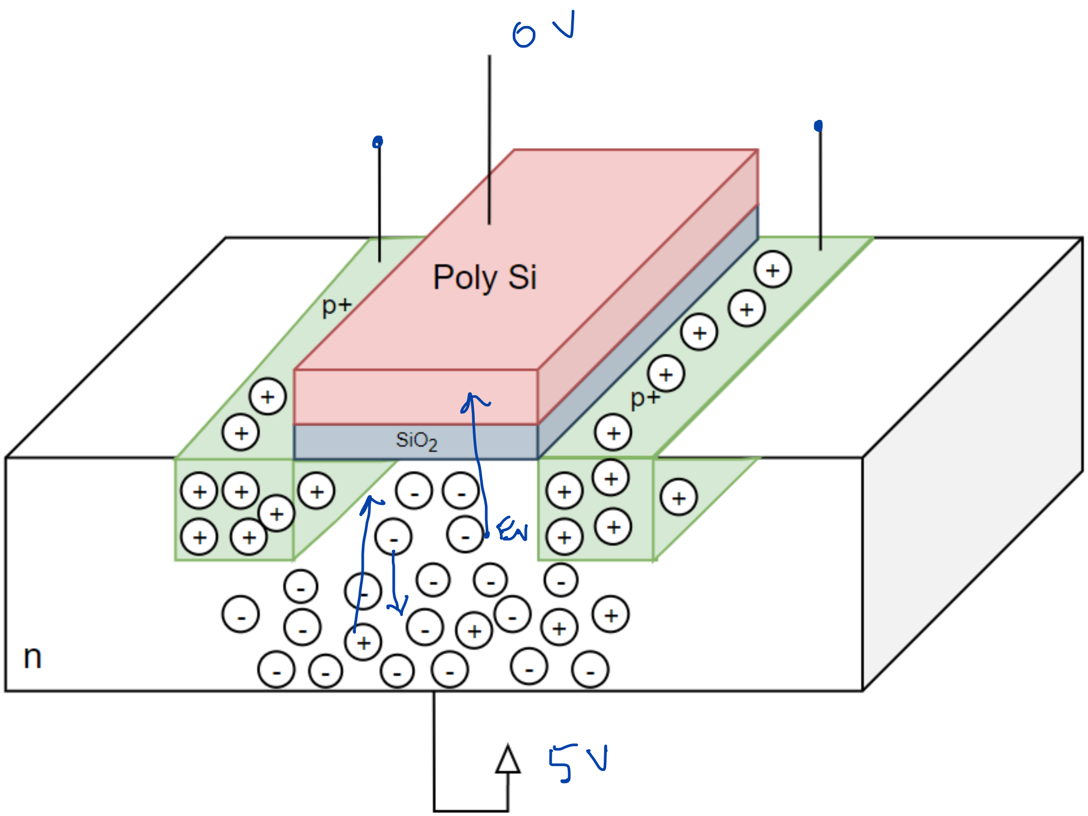


Carriers in p-transistor



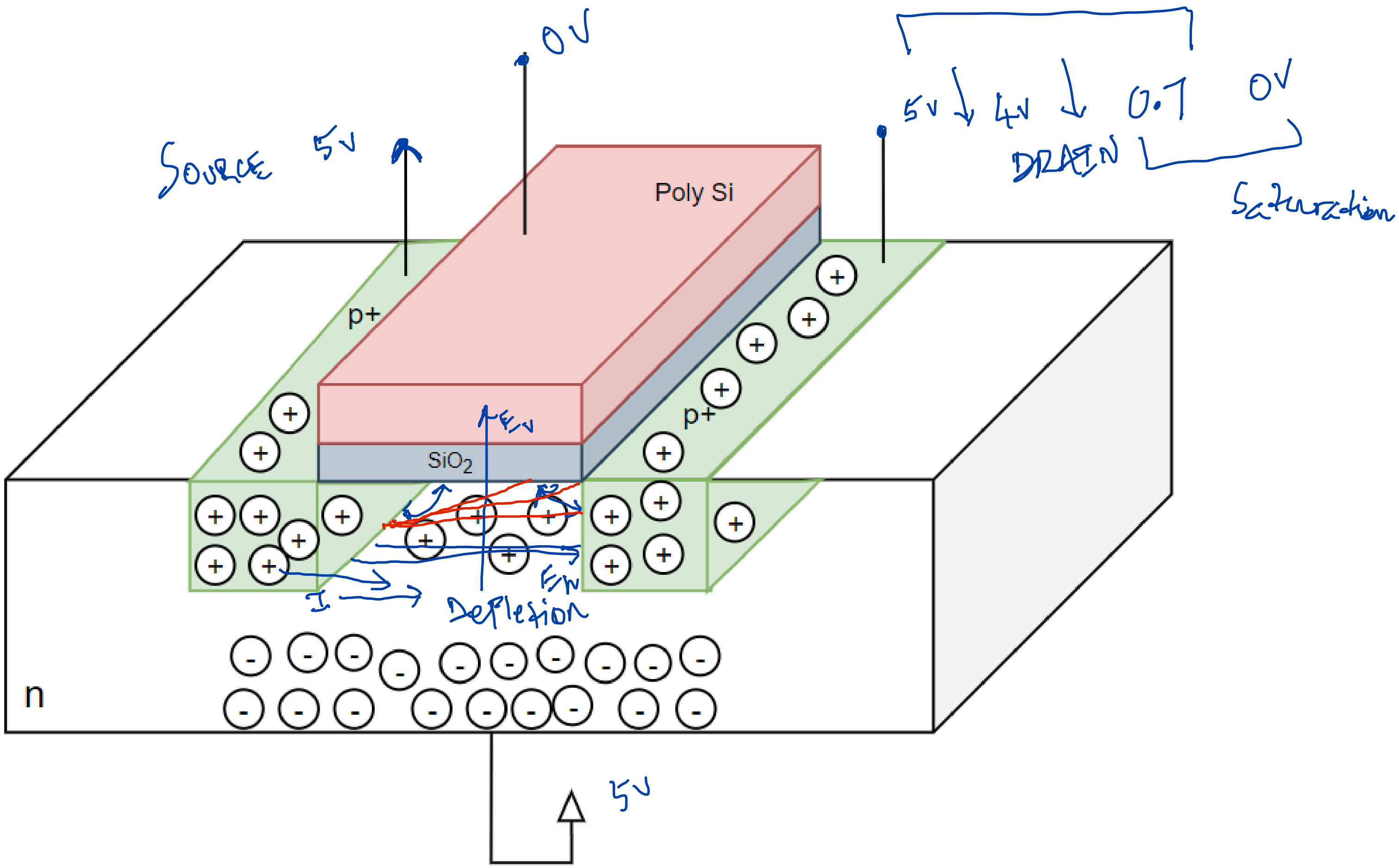
p-transistor as normally-closed switch

$V_{tp} = -0.7v.$

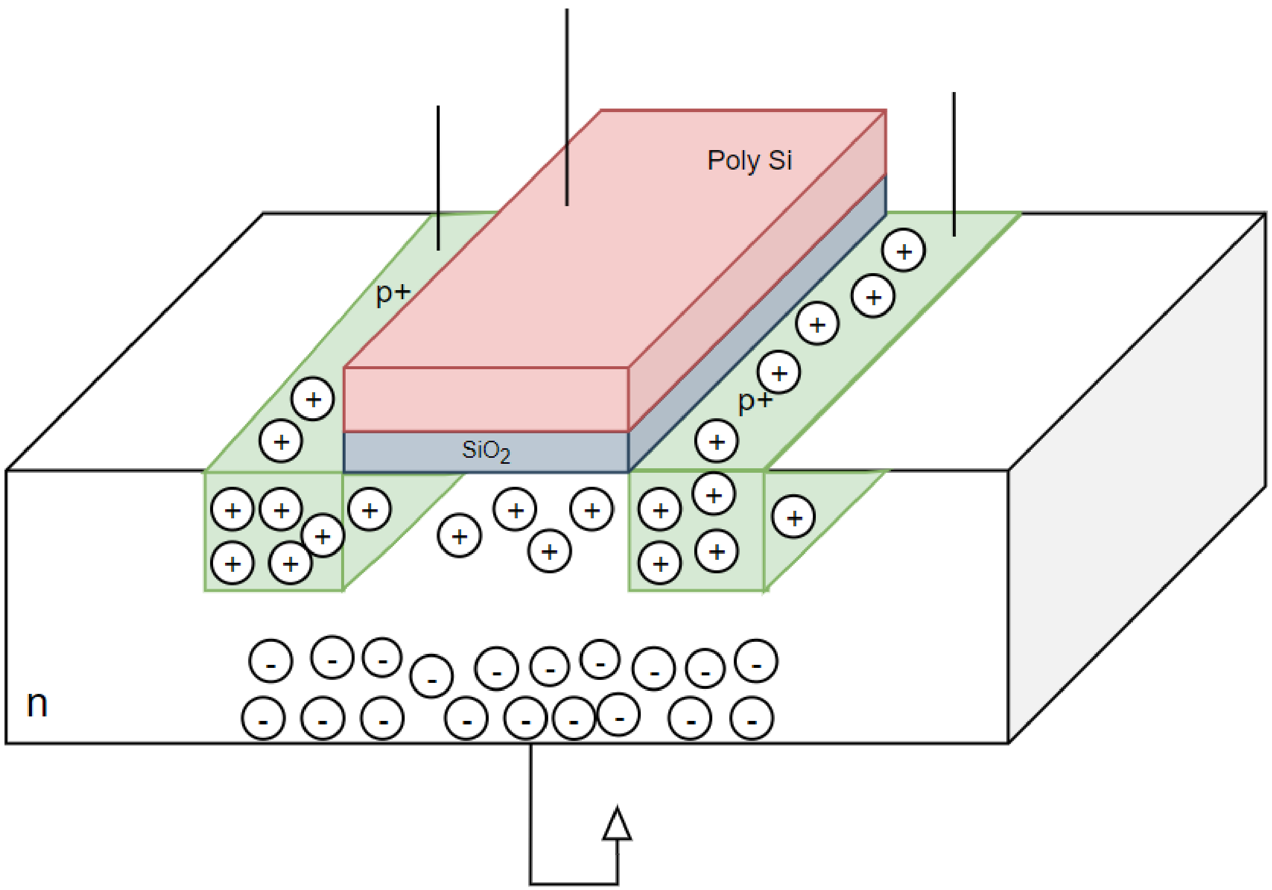


Conduction

Linear
Resistive

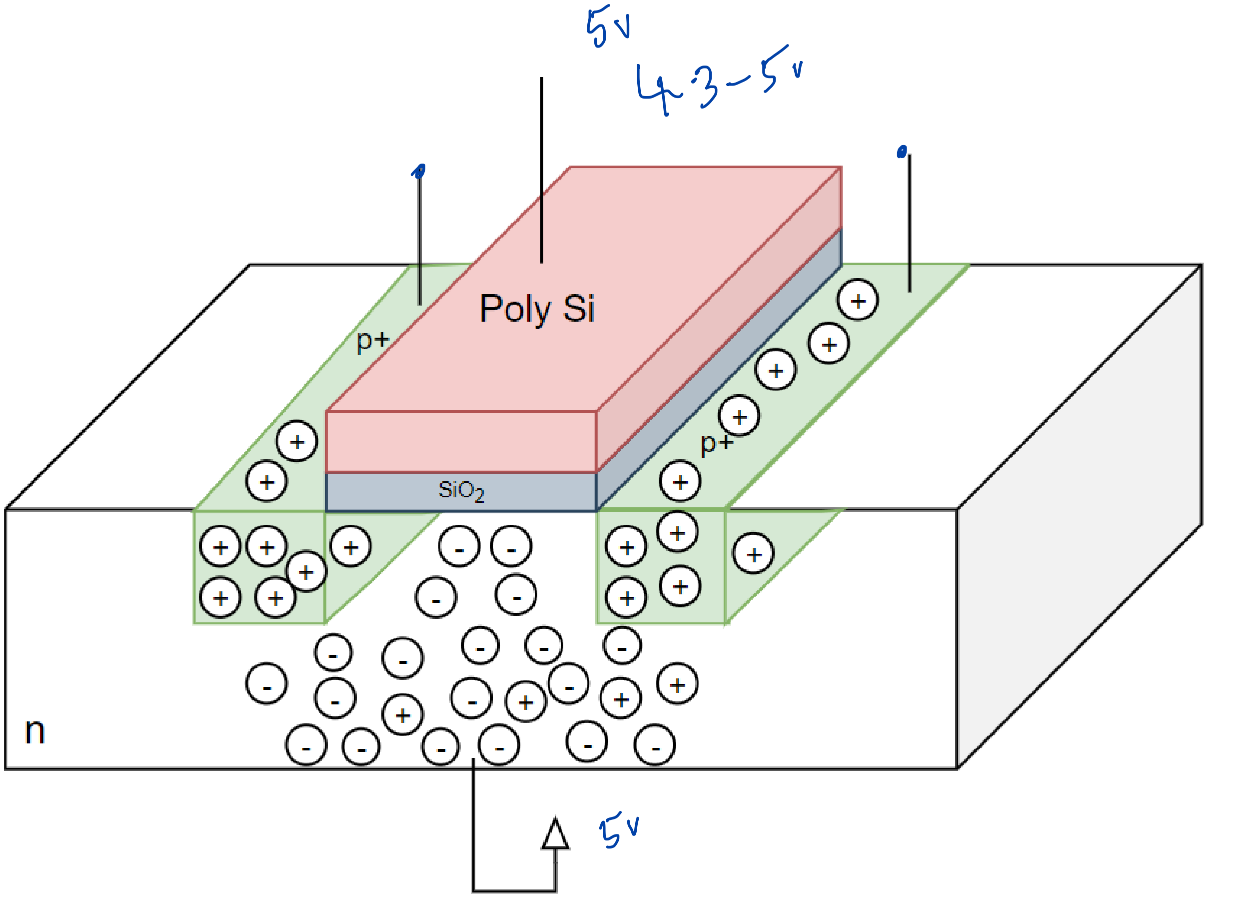


Saturation

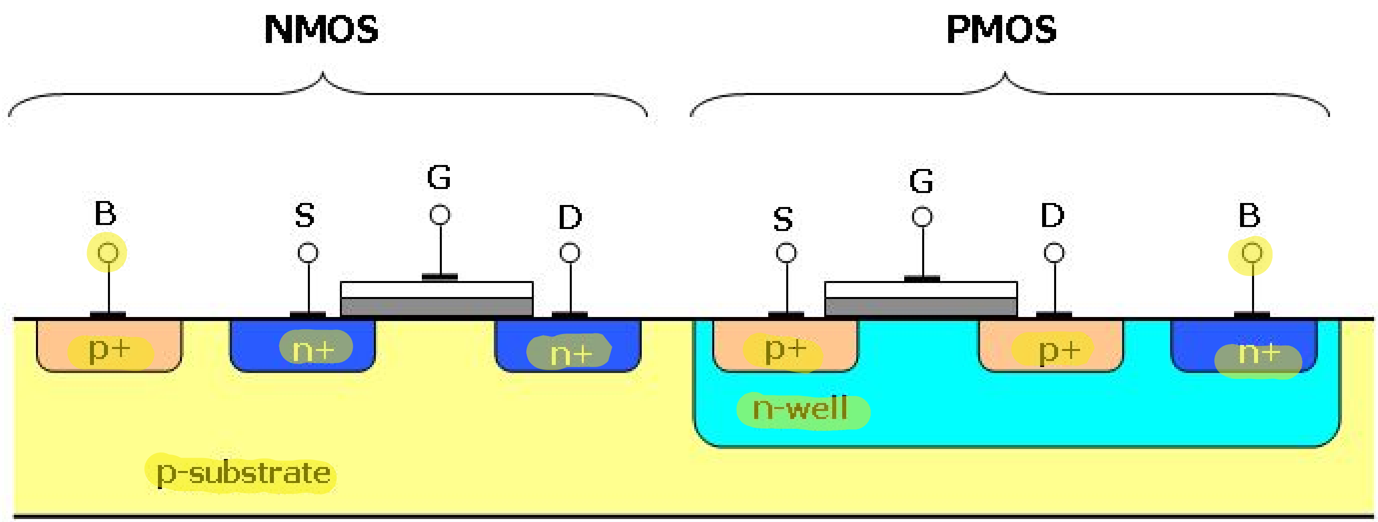


Cutoff

Open Switch



CMOS: n and p transistors

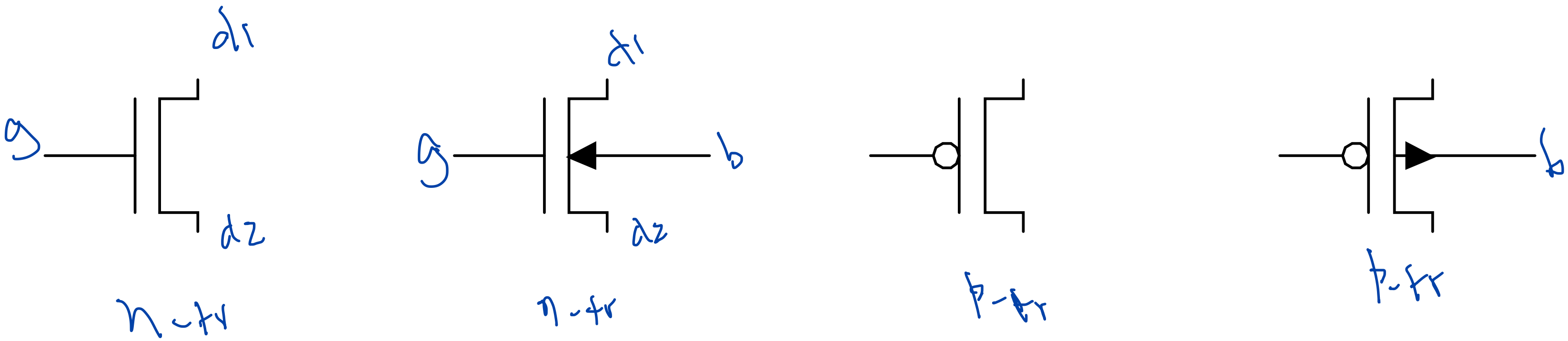


Latch-up

n-well CMOS process

p-well " "
Twin-Tub " "

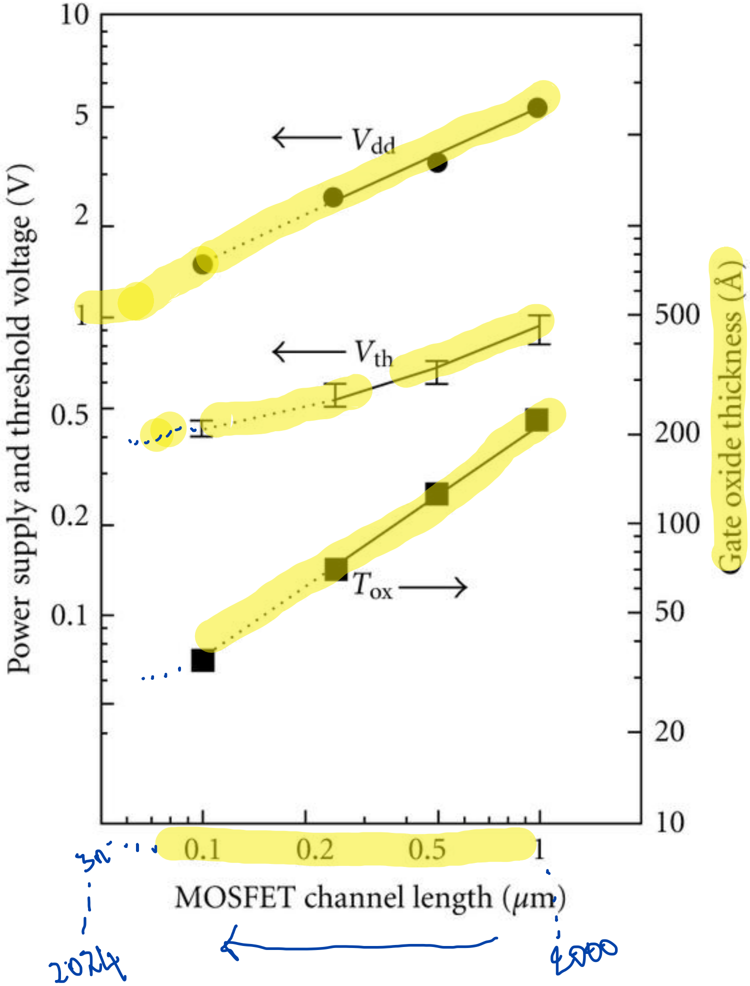
Symbols

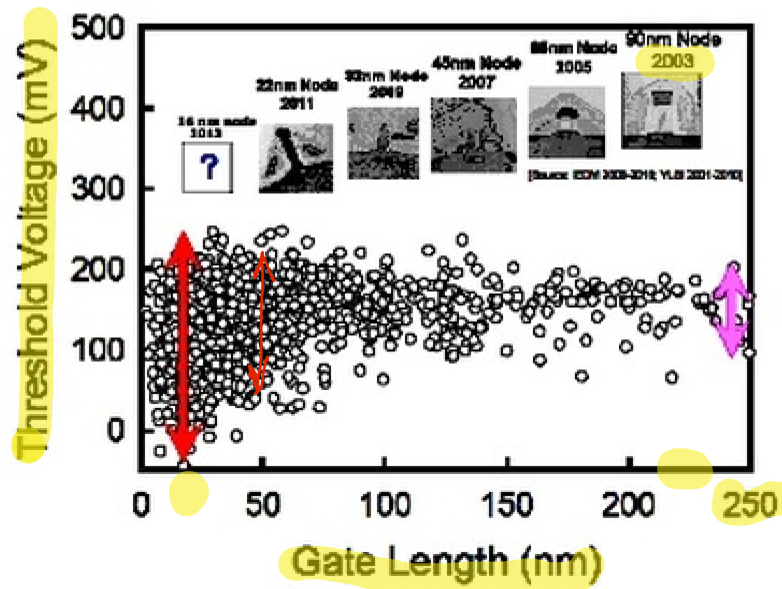


Summary of Operation

- n-transistor
  - Gate should above base by a threshold for inversion.
  - Linear mode: gate is above both diffusion ends by a threshold value.
  - Saturation mode: gate is above only one diffusion end by a threshold value.
- p-transistor
  - Gate should below base by a threshold for inversion.
  - Linear mode: gate is below both diffusion ends by a threshold value.
  - Saturation mode: gate is below only one diffusion end by a threshold value.

Supply and Threshold Voltages





Courtesy of Prof. Yiming Li, National Chiao Tung University, Taiwan

Figure 1. Simulation of threshold voltage variations vs gate length, showing the increasing range as devices are scaled down.